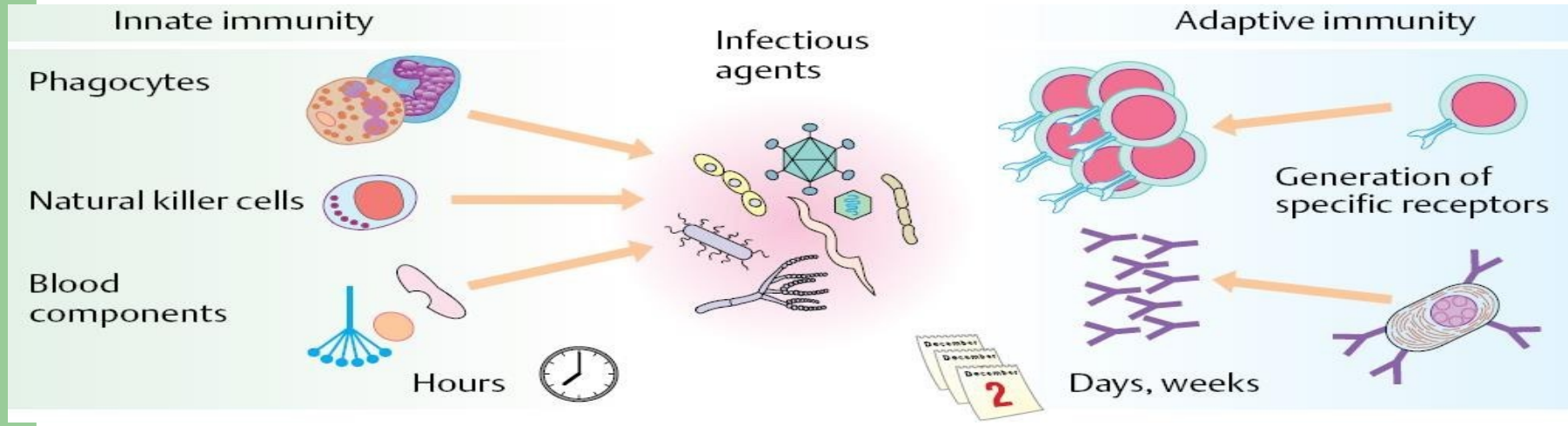
Innate immunity

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Types of Immunity:-

- There are two types of immunity:
- 1-Innate immunity.
- 2-Adaptive immunity.



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1-Innate Immunity (non-specific immunity):-

- Defense mechanisms are present before exposure to infectious microbes.
- Dose not improves after exposure to foreign antigen.
- Have no memory.
- No MHC (Major Histocompatibility Complex) restricted.
- Have receptors called **pattern recognition receptors** (PRR)

Important factors that are part of innate immunity:

- Mechanical factors/physical factor.
- Chemical factors.
- Biological factors.

A-Mechanical/physical factors:-

- **1-*Keratin layer of intact skin*** is **indigestible** by most microorganisms. But when the skin is damaged as in burns, traumatic injury or surgery, infectious can be a serious problem.

- ***2-Intact mucosal lining*** of different organs acts as a **protective barrier** to block the **adherenc**e of bacteria to epithelial cells.

A-Mechanical/physical factors:-

- **3-*Washing action*** of tears, saliva and urine.
- **4-*Respiratory cilia*** **expels** foreign substances including microbes.
- **5-*Coughing and sneezing reflexes*** expel **trapped foreign** substances on the mucous lining of the respiratory tract.

B- Chemical factors:-

- **1-*Fatty acids*** of ***skin, acid pH of sweat and sebaceous secretions*** inhibit growth of microorganisms, due to their bactericidal effect.

- ***2-Lysozyme in tears, nasal secretions and saliva*** degrades peptidoglycan an essential element present in bacterial cell wall.

B- Chemical factors:-

- **3-*Spermine and zinc*** in the semen are bactericidal effect.
- **4-*Lactoperoxidase*** in milk has bactericidal action.
- **5-*Low pH*** (acidic environment) of stomach and vagina inhibits growth of many microbes.

C-Biological factors:-

- **1-*Normal bacterial flora*** present in the throat, colon and vagina **inhibits colonization** by other pathogens (bacterial interference).
- The normal bacterial flora of the skin may produce various antimicrobial substances such as bacteriocines and acids.
- The normal flora competes with potential pathogens for essential nutrients.

C-Biological factors:-

2 ***Phagocytosis*** by neutrophils and macrophages.

3 ***Nutural killer*** (NK) cells are involved in extra- cellular killing of target.

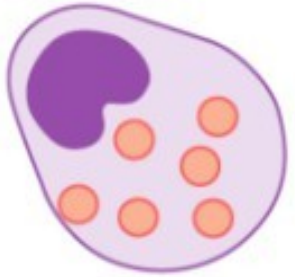
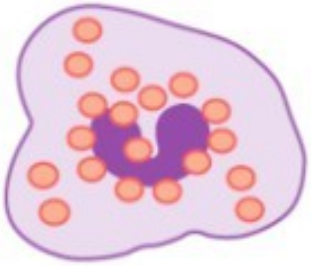
4 ***Eosinophils*** involved in killing of large parasites (helminthes).

5 ***Interferon*** some time secreted by virus infected cells inhibits viral replication.

C-Biological factors:-

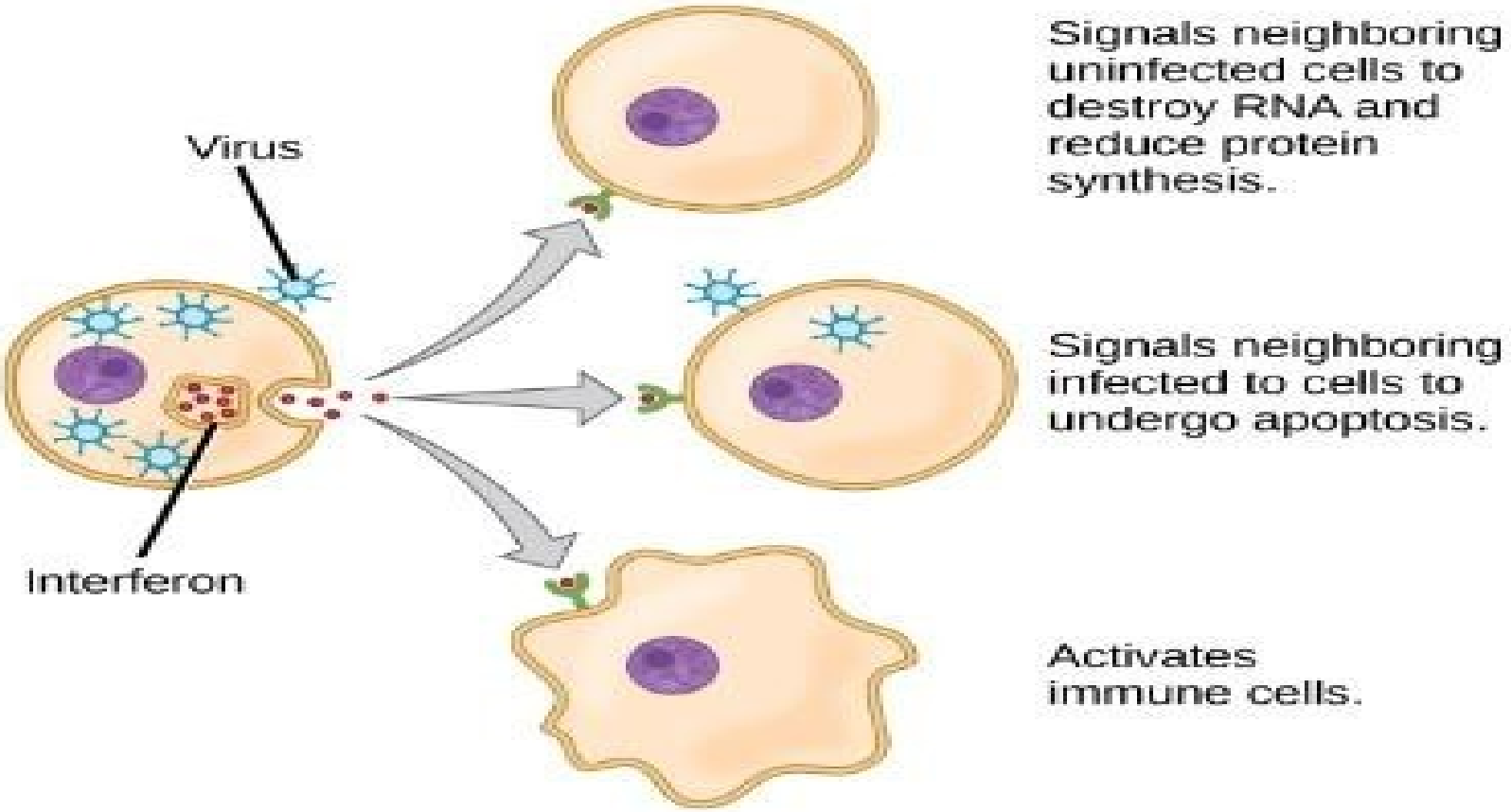
- **6-*Complements*** are plasma proteins that mediate cell lysis and inflammatory reaction.

- ***7-Acute phase proteins*** such as C-reactive protein (CRP), defencins, mannan binding lectin, Apolipoprotien B, serum amyloid protein, C9, factor B and fibrinogen.
- All act either to limit the spread of the infectious agents or to stimulate the host

Cell Type	Characteristics	Location	Diagram
Mast cells	Dilate blood vessels and induce inflammation through release of histamines and heparin. Recruit macrophages and neutrophils. Involved in wound healing and defense against pathogens, but can also be responsible for allergic reactions.	Connective tissues, mucous membranes	
Macrophages	Phagocytic cells that consume foreign pathogens and cancer cells. Stimulate response of other immune cells.	Migrate from blood vessels into tissues	
Natural killer cells	Kill tumor cells and virus-infected cells	Circulate in blood and migrate into tissues	
Dendritic cells	Present antigens on their surface, thereby triggering adaptive immunity	Present in tissues in epithelial tissue, including skin, lung and tissues of the digestive tract. Migrate to lymph nodes upon activation.	

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Monocytes	Differentiate into macrophages and dendritic cells in response to inflammation.	Stored in the spleen, move through blood vessels to infected tissues.	
Neutrophils	First responders at the site of infection or trauma, these abundant phagocytic cells represent 50–60% of all leukocytes. Release toxins that kill or inhibit bacteria and fungi and recruit other immune cells to the site of infection.	Migrate from blood vessels into tissues	
Basophils	Responsible for defense against parasites. Release histamines that cause inflammation and may be responsible for allergic reactions.	Circulate in blood and migrate to tissues	
Eosinophils	Release toxins that kill bacteria and parasites but also cause tissue damage.	Circulate in blood and migrate to tissues	


Virus
Interferon
Signals neighboring uninfected cells to destroy RNA and reduce protein synthesis.
Signals neighboring infected to cells to undergo apoptosis.
Activates immune cells.

Pattern-recognition receptor (PRR):

- It is a component of the innate arm that recognize what is foreign by detecting certain molecules on the surface of microorganisms that are **differ** from those in human cells.
- When a pathogen enters the body, cells in the blood and lymph detect the specific **pathogen-associated molecular patterns (PAMPs)** on the pathogen's surface.
- PAMPs are **carbohydrate, polypeptide, and nucleic acid** that are expressed by **viruses, bacteria, and parasites** but which **differ** from molecules on **host cells**.
- The receptors that recognize these PAMPs called pattern recognition receptors(PRR).

Pattern-recognition receptor (PRR):

There are two examples on PRR:

1- Toll-like receptors:

- Toll-like receptors (TLRs) are a family of transmembrane type of **PRR** that recognizes molecules that are shared by pathogens but distinguishable from host molecules that are broadly expressed by both **non-hematopoietic and hematopoietic cells.** The TLRs include TLR 1 to TLR 13.
- In mammals, TLRs are present on **various immune cells**, including natural killer cells, DC subsets, monocytes, macrophages, neutrophils, T lymphocytes and B lymphocytes.
- Some non-hematopoietic **epithelial, endothelial and organ parenchymal cells** also express these molecules.

Pattern-recognition receptor (PRR):

- Endotoxin (LPS) found on the surface of Gram-negative bacteria.
- LPS binds with protein present normally in the plasma called LPS-binding protein.
- This binding protein transfer LPS to a receptor on the surface of macrophage called CD14.

Pattern-recognition receptor (PRR):

- LPS stimulates a pattern recognition receptor called **toll-like receptor** 4(**TLR4**), which transmits the signal to the nucleus of the cell as a result cytokines will be produced.
- **TLR2**, signal the presence of **Gram-positive bacteria and yeast**.

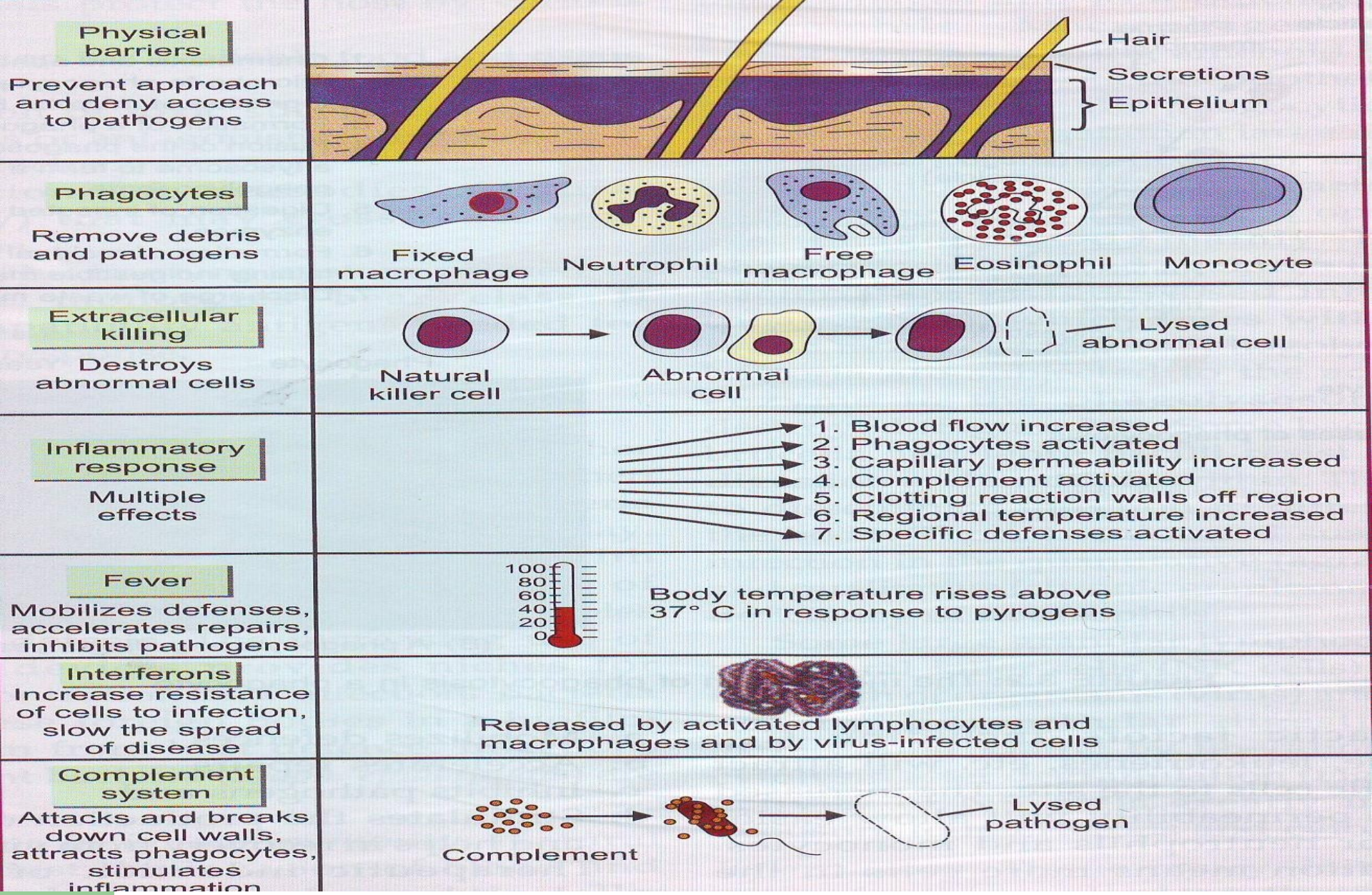
- TLR1 (functioning in TLR1/2 heterodimers),
- TLR2 (receptor for glycolipids, lipoteichoic acid, bacterial lipoprotein and components of mycobacterial walls)
- TLR4 (receptor for LPS, respiratory syncytial virus F protein, glycoinositolphospholipids, heat shock proteins and fibrinogen)
- TLR5 (receptor for bacterial flagellar protein)
- TLR6 (functioning in TLR1/2 heterodimers)
- TLR10 (receptor for undefined PAMPs)
- TLR11 (receptor for uropathogenic bacteria and their products as well as *Toxoplasma gondii*) are cell surface receptors that primarily recognize bacterial structures
- TLR3 (receptor for double-stranded RNA and synthetic polysaccharides)
- TLR7 (receptor for single-stranded RNA)
- TLR8 (receptor for single-stranded RNA and imidazoquinolone, an antivirus drug)
- TLR9 (receptor for unmethylated cytosine-phosphorothioate-guanine oligodeoxynucleotides (CpG-ODN)) are mainly localized intracellularly and recognize nucleic acids of microbial or viral origin.

Pattern-recognition receptor (PRR):

- **2-Mannan binding lectin (MBL)or mannose-binding protein:**
- **Many bacteria and yeast have a polysaccharide called mannan on their surface that is not present on human cells.**
- **MBL bind to the mannan on the surface of the microbes, which then activate complement resulting in the death of the microbe.**
- **MBL also enhances phagocytosis(acts as an opsonin) via receptors to which it binds on the surface of the macrophage.**

Pattern-recognition receptor (PRR):

- MBL also enhances phagocytosis(acts as an opsonin) via receptors to which it bind on the surface of macrophage.



Mechanisms of innate immunity (Mohanty,2006)

Summary

- There are 2 types of immunity innate and adaptive.
- Innate immunity includes mechanical, chemical, and biological factors.
- PRR is a components of the innate arm recognize what is foreign by detecting certain carbohydrates or lipids on the surface of microorganisms that are different from those in human cells.

References

- *Text book of immunology, edited by SK Mohanty and K Sai Leela; 2006.India.*
- *Medical microbiology and immunology, edited by Levinson W.* ©2006, Lange.
- *Cellular and molecular immunology , edited by Abbas A.K.; Lichtman A.H. and Pillai S.6th edition* © 2007,USA.
- Clinical immunology, edited by Chapel H. 5th edition © 2010, UK.
- Internet access.

Thanks a lot